

Program 9:

Organize Kubernetes Pods using Labels and Namespaces

19/12/2025

Project Directory:

```
prog9
|- namespaces.yaml
|- pods.yaml
```

Steps for Execution:

Step 1: Move to the directory where we work for DevOps lab. Then create a project directory with the name prog9

```
> mkdir prog9
```

```
> cd prog9/
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~$ cd devops_lab/
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ mkdir prog9
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ cd prog9/
```

Step 2: Write two files inside the prog9 folder,

1. namespaces.yaml 2. pods.yaml

The content of both the file is given in below pictures

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ nano namespaces.yaml
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ cat namespaces.yaml
```

```
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ nano pods.yaml
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ cat pods.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
  env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
  env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
  env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
  env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

Step 3: The project folder should look like the below image

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ ls
namespaces.yaml  pods.yaml
```

Step 4: Start the minikube if it is not started and also check the status by using the following command,

> minikube start --driver=docker --force

> minikube status

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ minikube start --driver=docker --force
🐳 minikube v1.37.0 on Ubuntu 24.04
! minikube skips various validations when --force is supplied; this may lead to unexpected behavior
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
   ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 5: Apply both the namespaces.yaml and pods.yaml files in the prog9 directory

> kubectl apply -f namespaces.yaml

> kubectl apply -f pods.yaml

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

Step 6: View all the pods in the namespaces by the commands,

> kubectl get namespaces

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ kubectl get namespaces
NAME                STATUS    AGE
default             Active   5m2s
dev                 Active   50s
kube-node-lease     Active   5m2s
kube-public         Active   5m2s
kube-system         Active   5m2s
prod                Active   50s
```

Step 6: Now verify the Pods by Namespaces using the commands,

> kubectl get pods -n dev

> kubectl get pods -n prod

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ kubectl get pods -n dev
kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           52s
frontend-pod  1/1     Running   0           52s
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           51s
frontend-pod  1/1     Running   0           52s
```

Step 7: Now Filter the pods by label using the commands,

- List all the pods in dev namespace
- > `kubectl get pods -n dev -l app=frontend`
- List all the pods in prod namespace
- > `kubectl get pods -n prod -l app=backend`

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           95s
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          101s
```

Step 8: Clean-up the space by deleting all the pods and namespaces by using the commands,

- > `kubectl delete pods --all -n dev`
- > `kubectl delete pods --all -n prod`
- > `kubectl delete namespaces dev prod`

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ kubectl delete pods --all -n dev
pod "backend-pod" deleted
pod "frontend-pod" deleted
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ kubectl delete pods --all -n prod
pod "backend-pod" deleted
pod "frontend-pod" deleted
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog9$ kubectl delete namespaces dev prod
namespace "dev" deleted
namespace "prod" deleted
```